

altairsim — altairsim machine examples

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Machines that boot. Each directory here is self-contained: a `.toml` that describes the machine, the media it needs lying beside it, and a note saying what you will see. Copy any one of them anywhere and it still runs — a path inside a machine file resolves against **that file**, not against the directory you launched from.

Each directory has its own `README.md` (also distributed as `README.pdf`) saying what the machine is and what to type. Each directory also holds a `<dir>.zip` containing all of its files, so you can download that one file instead of the files one by one. You do not need both. Launch a machine by naming its machine file from inside its directory:

```
cd examples/acr
altairsim mitstapes.toml
```

These machines need the [altairsim](#) simulator.

Disk systems

	What it is
<code>diskbasic/</code>	Altair Disk Extended BASIC 4.1 (MITS, 1977) on an 8" floppy behind an 88-DCDD, booted by the DBL PROM. Unlike the cassette BASIC, this one has a filesystem — <code>SAVE</code> by name, a directory, <code>DSKINI</code> . <code>MEMORY SIZE?</code>
<code>hdsk/</code>	CP/M 2.2 booting off an Altair 88-HDSK "Datakeeper" hard disk — an outboard controller moving whole sectors over an 88-4PIO — loaded by the HDBL PROM. <code>A0></code>
<code>fdcplus/</code>	CP/M 2.2b on an FDC+ : from a serial drive server (<code>fdcplus-type7.toml</code>) or on the FDC+'s 1.5 MB floppy (<code>fdcplus-type5.toml</code>).
<code>turnkey/</code>	The MITS 8800bt , an Altair with a Turnkey Module and no front panel, booting CP/M two ways: off an 88-DCDD floppy (<code>floppy.toml</code>) or an 88-HDSK hard disk (<code>hdsk.toml</code>).
<code>tarbell/</code>	CP/M 2.2 that boots itself off a Tarbell floppy controller and its own 32-byte boot PROM: single density (<code>tarbell.toml</code> , <code>tarbell #1011</code>) and double density, with and without DMA (<code>tarbelldd.toml</code> , <code>tarbelldd-dma.toml</code>).
<code>icom/</code>	CP/M and FDOS on an iCOM FD3712/FD3812 8" floppy — single and double density CP/M, and both revisions of iCOM's FDOS.
<code>cdos/</code>	CDOS 2.58 , Cromemco's CP/M work-alike, booted from an 8" diskette through a Cromemco 16FDC on a 4 MHz Z80.
<code>sdsys/</code>	The SD Systems SBC-200 , a Z80 single-board computer with the SD/MS monitor, booting SDOS, CP/M 2.2, and banked and non-banked CP/M 3 — with a serial or a video console.
<code>dualide/</code>	CP/M 3 off a CompactFlash card on the IDE half of the S100Computers IDE-AB CF+ESP32 board.
<code>dualsd/</code>	CP/M 3 off microSD cards on the S100Computers Dual SD board, booted from the V2 Z80 board's MASTER monitor.
<code>dualidesd/</code>	The whole IDE-AB CF + Dual SD combination board: all four drives, <code>A: / B:</code> on CF and <code>C: / D:</code> on SD, under CP/M 3.

Tape, storage and I/O boards

	What it is
acr/	Mike Douglas's MIT'S Tapes CP/M disk and WRTAPE , the utility that writes any of the MIT'S distribution BASICs back out through an 88-ACR — a bootable audio cassette you can then load on <code>basic4k</code> / <code>basic8k</code> / <code>ps2</code> , or on real hardware.
uio/	The 88-UIO , two cards in one — console and cassette on a single board — running Altair 8K BASIC 3.2 .
io4/	The SSM IO-4 (2P + 2S) as a console, running the SSM 8080 System Monitor. Explains the console straps.
pb1/	The SSM PB1 2708/2716 EPROM programmer: SSM's own burner programs from the manual, and the burned chip read back as Intel HEX.
pmmi/	The PMMI MM-103 modem, with a small 8080 program that turns it into a terminal.
printing/	An 88-C700 line printer , and a banner program that prints through it. The README sets up a real printer on your host — a network printer over <code>socket:</code> , or a CUPS queue over <code>printer:</code> — and a page comes out. Per-OS host setup (macOS, Linux; Windows pending).

Graphics and games

	What it is
dazzler/	A Cromemco Dazzler , 64×64 in 16 colors: Li-Chen Wang's Kaleidoscope , a pattern that wanders and recolors forever, DZMBASIC , Microsoft BASIC with <code>DZ0P</code> graphics commands, under CP/M, the Dazzler games with joysticks and sound on a Cromemco D+7A, and Cromemco's GDEMO graphics demonstration.
cadzilla/	The CADzilla graphics board's HD63484 ACRTC: <code>DRAWDemo</code> shows every drawing command, one screen each, under CP/M.
sol-20/	A Processor Technology Sol-20 with 1977 cassette games: TREK80, ATC, PAC-MAN and RAIDERS, each loading itself from its tape.

Sound

	What it is
newtech/	A Newtech Model 6 Music Board , a 6-bit D/A converter and a speaker: Newtech's own player, MICROPLAY , plays the score from the manual of the board, Scott Joplin's " The Entertainer ".